

Albuterol

(Proventil, Ventolin)

I. Classification

- Bronchodilator

II. Action(s)

- Selective B2-agonist – relaxes bronchial smooth muscle, causing bronchodilation

III. Indications

- Asthma / COPD
- Moderate to severe allergic reaction and anaphylaxis
- Other Condition with Bronchospasm / Wheeze (Drowning, CO Poisoning, Bronchiolitis)
- Suspected Hyperkalemia (Bradycardia & History Renal Failure, Chemotherapy)

IV. Contraindications

- Known hypersensitivity / allergy

V. Precautions

- The patient's vital signs should be monitored for tachycardia and hypertension
- Caution in elderly patients and those with cardiovascular disease or Hypertension – monitor closely for tachydysrhythmias.
- Lung sounds should be auscultated before and after treatment to determine efficacy of treatment.

VI. Drug interactions

- Beta-blockers may antagonize effect
- Ipratropium bromide increases duration of bronchodilation
- Caution if patient takes MAOIs, tricyclic antidepressants, sympathomimetics

VII. Adult Dosing and Administration

- 2.5 mg / 3 mL nebulized
- Suspected Hyperkalemia – 7.5 mg / 9 mL nebulized
- Repeat as needed and per guidelines

VIII. Pediatric Dosing and Administration

- 2.5 mg / 3 mL nebulized
- Repeat as needed and per guidelines

IX. Onset

- 5-15 minutes, Peak effects 1-1.5 hours

X. Duration

- 5 hours

XI. Possible adverse effects

- Hypersensitivity / allergy
- Paradoxical bronchospasm
- Tachycardia / hypertension / palpitations / arrhythmias / myocardial infarction
- Cough
- Nausea / vomiting
- Dizziness / nervousness / tremulousness